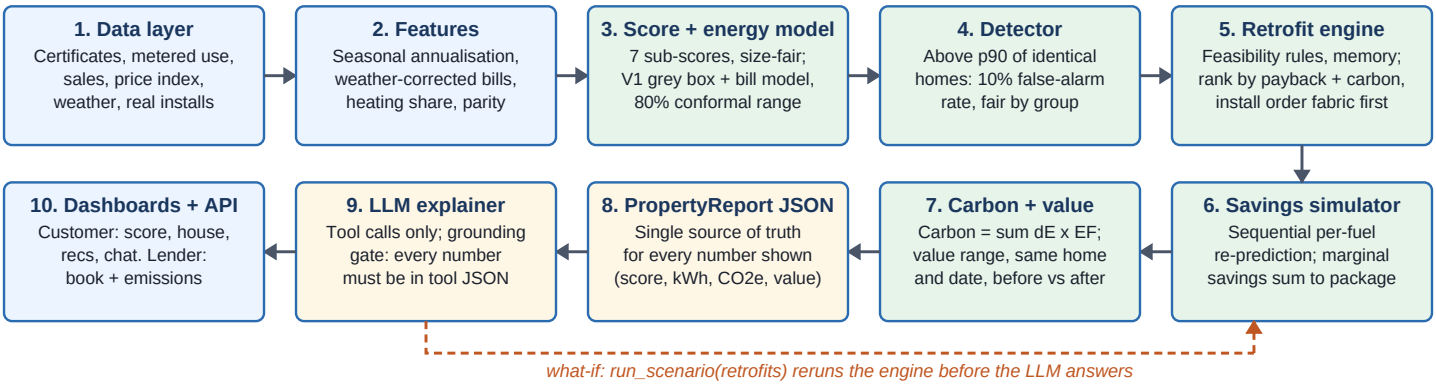
Green Home Score & Retrofit Recommendation Engine

A deterministic engine that scores a home, recommends feasible retrofits in the right order and estimates savings, carbon and value, validated on real metered homes and real installs, with an assistant that can only explain the engine's numbers

1. Solution

For each home the engine builds one **PropertyReport**, the single source of truth that the customer dashboard, the lender portfolio view, the API and the assistant all read and never recompute: a 0-100 Green Home Score with its seven weighted parts, expected electricity and gas use with an 80% range, a 'worse than similar homes' check, feasible retrofits in priority order (fabric, controls, heating, generation) with marginal per-fuel savings, bill saving and payback, carbon from official 2026 factors, and an indicative property value range before vs after. It runs on **542,711 real homes** from energy performance certificates in three contrasting local authorities (urban terraces, rural off-gas cottages, inner-city flats), joined to 175,457 real sales (92.2% match, only certificates lodged before each sale). Models learn from a 50,000-home national sample of **metered** annual gas and electricity, and are checked against **4,791 real loft, cavity-wall and solar installs** with metered use two years before and after. Synthetic data fills only what no public source has (monthly bills, customer IDs, smart controls, loan-to-value) and is never used to report accuracy.



2. Approach: What Is New

- **A grey-box energy model calibrated on real retrofits, not on its own equations.** V1 computes heat loss per element (roof, walls, windows, floor, air) times regional degree days, then log(use) = β · log(physics) + GBDT(home description). Steady-state physics over-predicts metered gas about 2× (the prebound effect: real homes are heated less than assumed), and over-predicts retrofit savings 3×. We fit the elasticity β = 0.38 on the observed installs in the train split only, so a retrofit's effect is (physics after / before)$^{\beta}$: smooth, never the wrong sign, with diminishing returns and interactions from the physics, and a per-element heat-loss share for the house diagram. The residual never sees the physics value: when it did, 18% of gas homes were told insulation raises use.
- **Non-circular evaluation.** Headline accuracy comes only from real metered homes the model never saw, on a fixed split with the **test set scored once**; every choice was made on 5-fold CV. Changes made after that are labelled post-test and judged on validation. Synthetic bills carry hidden occupancy and behaviour the engine never sees, and are used only to test the pipeline and the parts real data can't (controls, heat pumps, glazing).
- **Two prediction modes, and a model that learns from the customer's own bills.** *Similar homes* (description only) for prospects and for the detector, which must not see the home's own use; *this home* for existing customers, a GBDT on the year-on-year change log(this year / last year). Since the bill alone carries most of the signal, it is the reference to beat. Each year of history is weather-corrected to a typical year with the home's own degree-day fit, as the national metered data is; the same fit splits each home's heating from hot water and cooking for the savings simulator.
- **Honest uncertainty with stated error rates.** Conformalised quantile models (CQR) give every prediction an 80% range. The 'worse than similar homes' flag fires above the 90th percentile for identical homes, so its false-alarm rate is 10% by construction, and it is checked in every region, type, age, rating and deprivation group. A big house is compared with similar big houses.
- **A size-fair score.** Raw kWh/m^2 scored small homes 23.5 points below large ones on real metered use, penalising flats and smaller (often lower-income) homes for their size. The consumption part now scores actual / expected for similar homes: the gap falls to 3.6 points. Every mapping is monotone and in config with a source.
- **A value uplift that compares homes on the same streets.** Without location controls the rating premium comes out **backwards** (badly rated homes are often expensive central period houses). A hedonic model with 259 postcode-sector fixed effects and clustered errors gives the expected order (vs D: A/B +1.0%, C +0.7%, E -2.7%,

F/G -5.1%), confirmed by a refit on test sales. The report shows the same home on the same date before vs after, as a range that can include 0, labelled an association, never causal.

- **Recommendations ranked by value and checked against hidden truth.** Suitability comes from payback at current capped prices, with a carbon floor and an 'insulate first' rule for heat pumps. The ranking is scored against the synthetic world's hidden true savings, not the engine's own numbers.
- **An assistant that cannot invent numbers.** The LLM only reads tool output (report, scenario rerun, customer memory). A what-if question reruns the engine before the model answers, and a grounding gate checks every number in the answer against the tool JSON, retries once, then falls back to a deterministic template. Customer memory turns stated constraints (tenant, declined measures) into feasibility rules.

3. Features

- **Customer view:** score with points per part; heat-loss house (3D and SVG) coloured by each element's share of heat loss, click a part to see its fix; what-if toggles that rerun the engine and move the score, savings and house together; ranked recommendation cards (bill saving, payback, carbon, cost, why); grants and green-finance matching with monthly cash flow; the grounded assistant with an audit of every number.
- **Lender view:** a mortgage book of real homes: score and rating mix, share Poor / Very poor, financed emissions (PCAF, loan-to-value attribution), reduction potential, a 2030 EPC C target simulator and a ranked list of who to contact about green finance.
- **Electricity and gas are tracked separately end to end;** a heat pump's extra electricity is a negative saving, and solar cuts what is bought from the grid, not demand. Missing data is 'unavailable' and flagged, never 0. FastAPI exposes reports, scenarios, portfolio, chat and memory.

4. Results

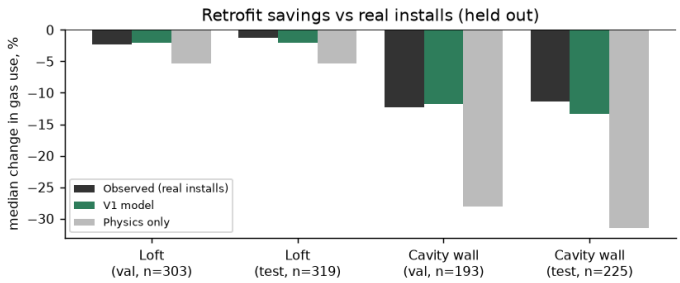
Energy model

Real metered homes, test split (scored once)	Gas MAE (kWh)	Gas median error	Gas R^2 (log)	Elec MAE (kWh)	Elec median error
Median of similar homes (no model)	4,177	30.7%	0.25	1,569	37.9%
Log-linear model	4,061	30.3%	0.28	1,573	38.6%
Physics x one scale factor	4,546	33.5%	0.14	-	-
V2: LightGBM, Optuna-tuned	4,008	29.4%	0.28	1,533	37.2%
V1 grey box (shipped), description only	**4,018**	**29.6%**	**0.27**	**1,535**	**37.2%**
This home: last year's bill (no model)	2,005	11.4%	0.67	917	16.0%
This home: change model with the bill	**1,953**	**10.9%**	**0.69**	**886**	**15.6%**

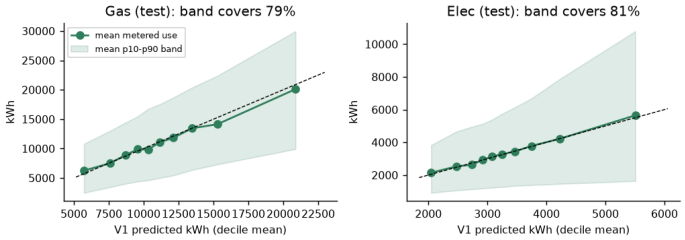
- **Single homes are noisy, books are not.** Identical homes vary 0.5-1.7× in gas use, which caps description-only accuracy (Optuna tuning gained 1.1% on gas; CatBoost and interaction features moved MAE < 0.3%, inside fold noise). Yet predicted totals for 1,000 homes are within **1.7%** of metered, gas R^2 is **0.79** across groups of similar homes, and the top 25% of users are found with precision 0.55 (0.25 random). That is what a lender's portfolio view needs.
- **80% ranges hold on new homes:** test coverage 79.2% gas, 80.7% electricity, and 76-84% in every region, type, age, rating and deprivation group. The detector flags 10.2% (gas) / 9.7% (electricity) of test homes against 10% expected, and 6-12% in every group.

Real installs, test split	Observed	Physics alone	V1	V2 (GBDT)
Cavity wall insulation (225 homes)	-11.4%	-31.5%	**-13.4%**	-3.8%
Loft insulation (319 homes)	-1.3%	-5.3%	**-2.1%**	no loft input

- **Savings match real installs.** One fitted β brings V1 within about 2 points of the observed change for both measures on held-out installs, where physics alone overstates 3-4×. The tuned GBDT sees only a third of the cavity saving: a model trained on consumption levels learns a weak between-home difference, not a before-after change. This, not accuracy (the two tie), is why V1 ships. The solar rule doubled real savings (1,260 vs 487 kWh/yr observed); recalibrated on train installs it predicts -488 vs -483 on validation.



Retrofit savings vs real installs, held-out splits: V1 tracks the observed change, physics alone overstates it 3-4x.



Test calibration by decile of V1's prediction: mean metered use follows the diagonal; shading is the mean 80% range.

- **Post-test, chosen on validation:** a second year of the customer's bills cuts error a further 1.5% (gas) and 2.7% (electricity); weather-correcting the history removes a 7% under-forecast after a mild year (synthetic check: error 798 → 308 kWh). On 1,175 synthetic mid-series retrofits covering all seven measures, the predicted total is within 3% of the hidden truth and a heat pump's electricity rise has the right sign every time.

Value, recommendations, assistant

- **Value (test):** median error **14.9%** vs 24.6% for the area x type median and 32.6% for the baseline in the brief; R^2 0.76. For homes sold before, 85% indexed last sale + 15% model gives 8.2%. 80% ranges cover 79.3%, and 78-81% in every area, type and rating band.
- **Recommendations:** against hidden true value per pound on 1,691 synthetic homes, NDCG@3 is **0.978** vs 0.813 for the brief's fixed-order baseline, and the top pick is right 91% vs 38%. A strong 'cheapest first' reference nearly ties on bill value (0.968), but the engine clearly wins on carbon per pound (0.968 vs 0.904).
- **Assistant:** on 50 held-out adversarial scenarios, all 42 traps (invented figures, wrong signs, infeasible asks) were blocked and all 8 legitimate answers passed: groundedness 100%, unsupported claims 0%. What-if answers on 20 real homes: 60 of 60 grounded after the engine rerun.

Honest limits

- V1 gets the *average* saving right, but real installs save about the same whatever physics predicts (cavity slope 0.01 on train vs 0.38 assumed), so per-home fabric savings for the worst homes are the optimistic end. Electricity is driven by occupants (R^2 0.16 from the description). Monthly bills for real homes don't exist publicly, so bill-based gains are shown on validation and synthetic data. Property value lacks a street-level input. Each is quantified in the evaluation docs.

Engineering. 186 automated tests (monotone score mappings, feasibility, per-fuel carbon, physics directions, feature parity between training and deployment, grounding). *make build* rebuilds the real data layer, *make train* and *make eval-test* train and score every model, and trained models and processed data install from hosted links in one command each (verified from a clean clone). Every constant in the configuration names its source.